SALMA TAREK ABDELMAGEED

Junior Data Scientist

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Data Science graduate of Alexandria University (2026, GPA 3.58/4.0) with hands-on experience in machine learning, data analysis, NLP, computer vision and end-to-end pipeline development. Built predictive models, statistical analyses and interactive dashboards, plus a full-stack AI platform (React, FastAPI, PostgreSQL) as a graduation project. Certified in Web Development (PHP) and Android Development. Motivated to turn raw data into actionable insights and solve business problems with data-driven approaches.

EDUCATION

Bachelor of Computers and Data Science (Major: Data Science) 2022 – 2026

Faculty of Computers and Data Science, Alexandria University GPA: 3.58 / 4.0

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C++, PHP, JavaScript, Kotlin

Data Analysis & Visualization: Microsoft Excel, Power BI, Pandas, Matplotlib, Seaborn, Plotly, EDA, Statistical Analysis, Dashboards & Reporting

Data Science & ML: Machine Learning, Regression, Classification, Clustering, EDA, Feature Engineering, Statistical Analysis, PCA, NLP, Computer Vision, RAG, LLM Integration

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, TensorFlow.js, OpenCV, MediaPipe, spaCy, LangGraph, Matplotlib, Seaborn, Plotly

Backend & Web: FastAPI, SQLAlchemy, REST APIs, JWT Authentication, Node.js, React, HTML, CSS

Databases: PostgreSQL, MySQL, MongoDB

Tools & Concepts: Git, GitHub, Jupyter Notebook, FFmpeg, OOP, Data Structures, Algorithms

KEY PROJECTS

Smart Sports: AI-Powered Sports Networking & Fitness Platform *Graduation Project*

Full-Stack Web App | Computer Vision, NLP & LLM Integration | React, FastAPI, PostgreSQL, MediaPipe, TensorFlow.js, spaCy

GitHub | Demo Video

- **Full-stack ownership:** built the React frontend (player, club and admin dashboards, dark mode, live camera) and the FastAPI backend (SQLAlchemy/PostgreSQL, JWT auth, bcrypt, role-based access).
- **Exercise video analysis:** MediaPipe Pose + OpenCV compute joint angles and count reps for 20 exercises with adaptive thresholds; FFmpeg merges the annotated skeleton video with audio.
- **Real-time webcam coach:** TensorFlow.js MoveNet runs in the browser for live pose tracking, good/bad-rep counting and instant form feedback with no server round-trip.
- **AI voice coach:** FastAPI connected to an LLM hosted on Kaggle/Colab (exposed via ngrok) returning spoken mp3 feedback and coaching text for each analyzed workout.
- **CV matching (NLP):** pdfplumber/python-docx + spaCy extract skills from player CVs and score them against each club's position requirements.
- **Nutrition & workout planner:** macro engine (rule-based + Keras model) and exercise planner generate 7-day plans stored per user as JSON.
- **Multi-agent chatbot:** LangGraph + Groq agents (medical RAG, guide, CV feedback, player analysis with Plotly radar chart) served as a separate service integrated into the React app.
- **Club–player workflow:** requests, offers, accept/reject, notifications, job board and admin analytics panel.

Retail Sales Analysis: Walmart Dataset

Exploratory Data Analysis | Python, Pandas, NumPy, Matplotlib, Seaborn

GitHub | Portfolio

- Analyzed 3 years of weekly sales data across 45 Walmart stores to identify revenue drivers, seasonality and holiday impact.
- Findings: Store 20 had the highest revenue (~$301M); Thanksgiving week outperformed the average by 15%; April–August was the strongest period.

Airline Passenger Satisfaction Prediction
Supervised Learning | Classification | Python, Scikit-learn, Matplotlib
GitHub | Portfolio

- Built and compared KNN and Gaussian Naive Bayes classifiers on 129,880 passenger records: KNN reached 92.7% accuracy (F1: 0.93) vs 86.4% for Naive Bayes.
- Key insight: online boarding drives satisfaction more than seat comfort.

US Airline Twitter Sentiment Analysis
NLP | Sentiment Analysis | Python, Pandas, Matplotlib
GitHub | Portfolio

- Analyzed 14,600+ tweets about 6 US airlines: 63% negative, 16% positive; "Customer Service Issue" was the top complaint driver.
- Identified sentiment gaps between airlines for competitive benchmarking.

Hotel Reservations & Cancellation Prediction
End-to-End ML | Clustering | Python, Scikit-learn, Plotly
GitHub | Portfolio

- Analyzed 36,000+ booking records and built predictive models (KNN: 80.6%, Logistic Regression: 79.8%); found that 1 in 3 bookings is cancelled.
- Applied K-Means clustering for customer segmentation and tracked revenue growth from $1.05M to $6.06M.

Student Career Readiness Prediction
Feature Selection | Dimensionality Reduction | Python, Scikit-learn
GitHub | Portfolio

- Applied PCA and Sequential Forward Feature Selection: 3 key features (Career Readiness %, Internship, CGPA) explain 95% of model decisions.
- Reduced survey complexity from 40+ questions to a small set of core predictors.

CERTIFICATIONS

Web Development with PHP — Information Technology Institute (ITI) 2025
Android Application Development — Route Academy 2024

LANGUAGES

Arabic: Native | English: Intermediate (B1–B2)

SOFT SKILLS

Problem Solving, Critical Thinking, Teamwork, Communication, Time Management, Adaptability, Self-Learning, Presentation